

PHASE 6

Performance Testing Phase

Functional, responsive, and embed-performance testing of the PulseView site

PulseView – Heart Disease Risk Analysis Dashboard

Project	PulseView – Heart Disease Risk Analysis Dashboard
Document	Performance Testing Report
Version	1.0
Date	July 2026
Prepared By	Project Team

1. Introduction

This document records the testing carried out before release: the overall strategy, the executed test cases, defects identified, and observations on the performance of the embedded Tableau views, which are the heaviest part of the page.

2. Test Strategy

Test Type	What it covers
Functional testing	Each route (/ , /about, /dashboard, /story, /contact) returns the correct page and content.
UI / navigation testing	Story tabs, previous/next buttons, and keyboard arrow-key navigation stay in sync.
Form validation testing	Contact form rejects incomplete submissions and confirms valid ones.
Responsive / cross-browser testing	Layout holds up across common breakpoints and current browsers.
Embed performance testing	Tableau dashboard and story iframes load and become interactive within an acceptable time.

3. Test Cases

ID	Module	Steps	Expected Result	Status
TC-01	Home	Navigate to /	Hero section loads with links to Dashboard and Story	Pass
TC-02	About	Navigate to /about	Problem statement, 3 personas, and 6-step process display correctly	Pass
TC-03	Dashboard	Navigate to /dashboard	Tableau dashboard iframe loads; all 4 panel descriptions render	Pass
TC-04	Dashboard	Resize browser to tablet width	Dashboard iframe and panel text reflow without overlap	Pass
TC-05	Story	Navigate to /story, click each tab	Active scene image + copy updates; progress label updates (e.g. 02 / 04)	Pass
TC-06	Story	Click Next/Prev buttons repeatedly	Navigation wraps correctly from last scene back to first and vice versa	Pass
TC-07	Story	Press Left/Right arrow keys	Scene changes match on-screen tab/button navigation	Pass
TC-08	Contact	Submit form with all fields empty	Flash error: "Please fill in every field before sending."	Pass
TC-09	Contact	Submit form with valid name, email, message	Flash success message including submitted name; fields reset	Pass

ID	Module	Steps	Expected Result	Status
TC-10	Contact	Submit form with only the email field filled	Flash error shown; form does not silently accept partial data	Pass
TC-11	Cross-browser	Load all 5 pages in Chrome, Firefox, Edge, Safari	Layout and embeds render consistently across browsers	Pass
TC-12	Performance	Measure time-to-interactive for /dashboard and /story	Embedded Tableau view becomes interactive within a few seconds on broadband	Pass

4. Defects & Resolutions Log

ID	Observation	Resolution / Status
OBS-01	Contact form has no email backend wired up, so submissions are not actually delivered anywhere.	Documented as a known, deliberate scope boundary (see Requirement Analysis, Section 7) rather than a defect; flagged as future work.
OBS-02	Tableau embed takes visibly longer to load than the rest of the page on slower connections.	Accepted: embed rendering is delegated to Tableau Public and is outside the Flask app's control. No fix required in this iteration.
OBS-03	Very narrow mobile widths compress the story tab labels.	Verified acceptable — labels remain legible; noted as a candidate refinement for a future responsive-design pass.

5. Performance Observations

- Static pages (Home, About, Contact) render effectively instantly, since they carry no external embeds — only local CSS/JS.
- The Dashboard and Story pages depend on the Tableau Public iframe for their heaviest content; their perceived load time tracks Tableau Public's own performance rather than the Flask app.
- No server-side database queries are involved in any route, so the Flask application layer itself introduces negligible latency.

6. Conclusion

All planned functional, navigation, form-validation, and cross-browser test cases passed. The two accepted observations (no email backend, Tableau-dependent load time) are scope decisions rather than defects, and are carried forward as noted future work in the final Project Documentation.